Evaluating Urban Traffic Congestion using **OpenStreetMap** and Sparse Public Transport Data

Andrei Bejan

(joint work with Richard Gibbens, David Evans, Alastair Beresford, Jean Bacon, Ken Moody and Adrian Friday)








DATA DESCRIPTION
JOURNEY EXTRACTION
JOURNEY TIME VARIABILITY ANALYSIS
INFLUENCE OF CONTEXTUAL FACTORS
ROAD TRAFFIC AND CONGESTION

The origins and main features of the data and tools used:

- bus locations recorded using GPS at the rate of ‘once every 20–30 seconds’ or more rare (+ ‘silence’ periods)
- source: archived and live bus location data stream (+ parsing and cleaning)
- road network and features: OpenStreetMap data
- bus stops info: NaPTAN data (UK National Public Transport Access Node database) and OSM
- bus routes: timetables and/or OSM



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- bus routes: timetables and/or OSM
- a GPS tracker



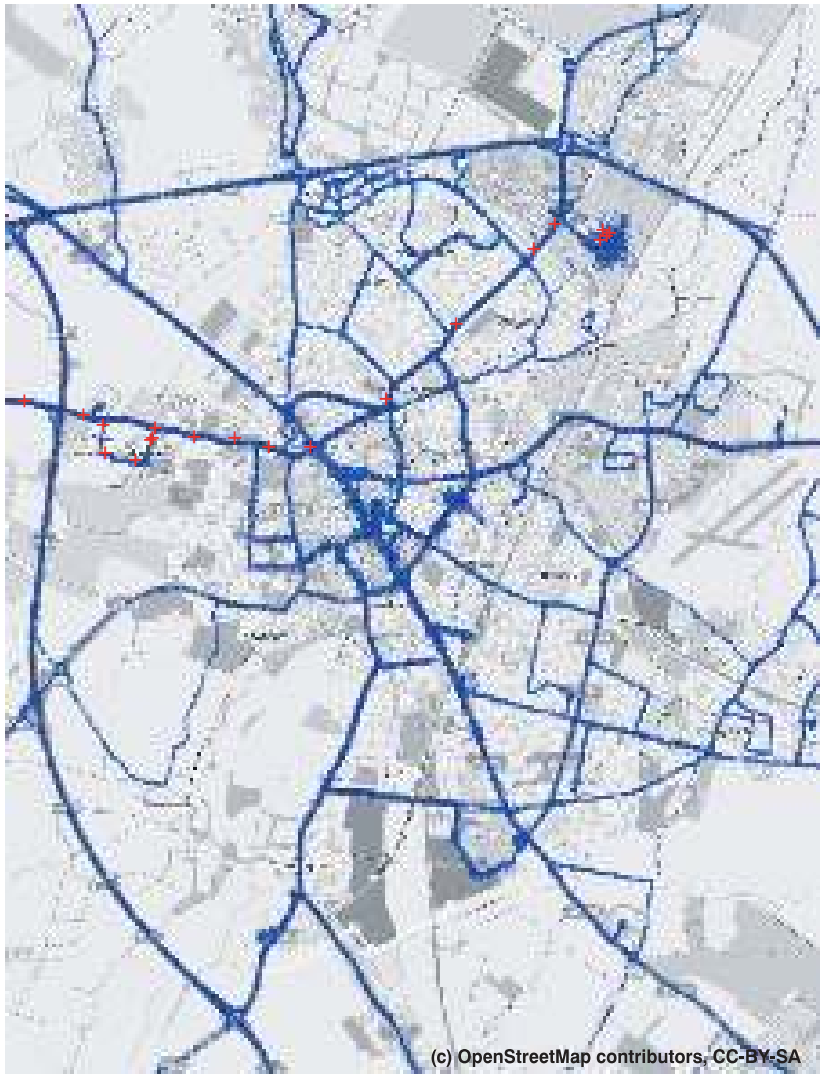
Road network coverage: each cell on the map corresponds to a 10 $\times$ 10 m square

Bus coverage of Cambridge ($\approx$ 150 buses)



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An example of a bus trajectory

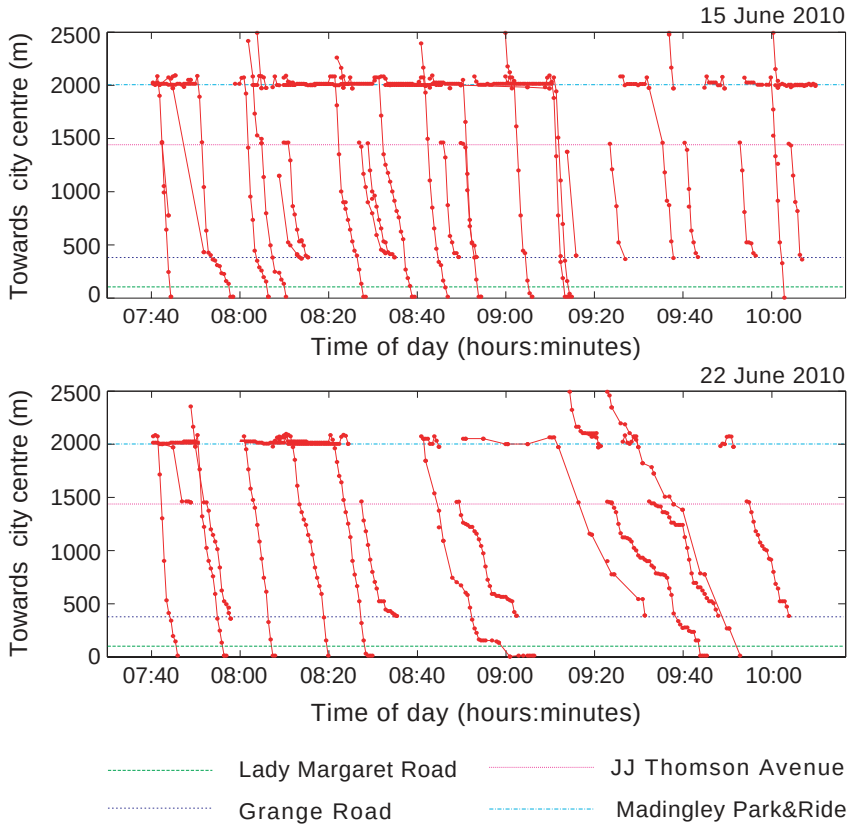


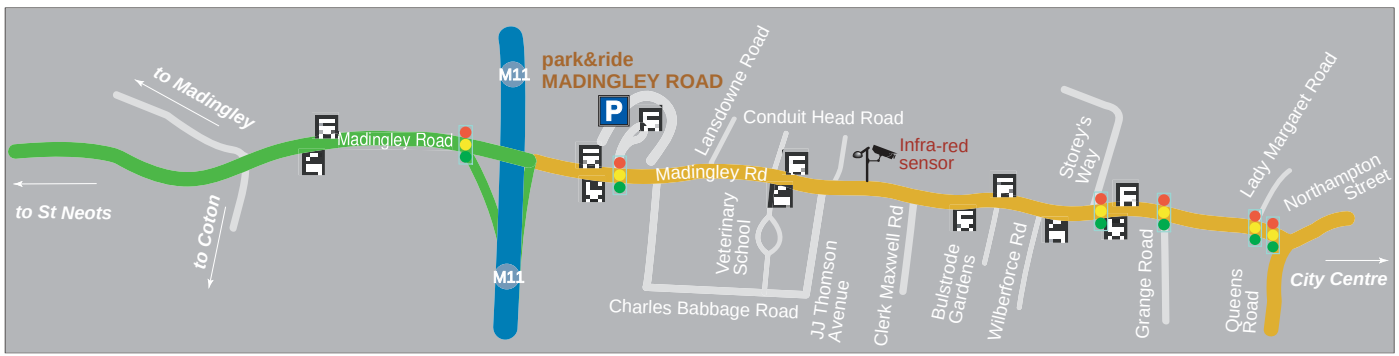
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Case studies: Madingley Rd, Histon Rd and A14 accident



Bus ‘trajectories’ in time: Madingley Road, Cambridge

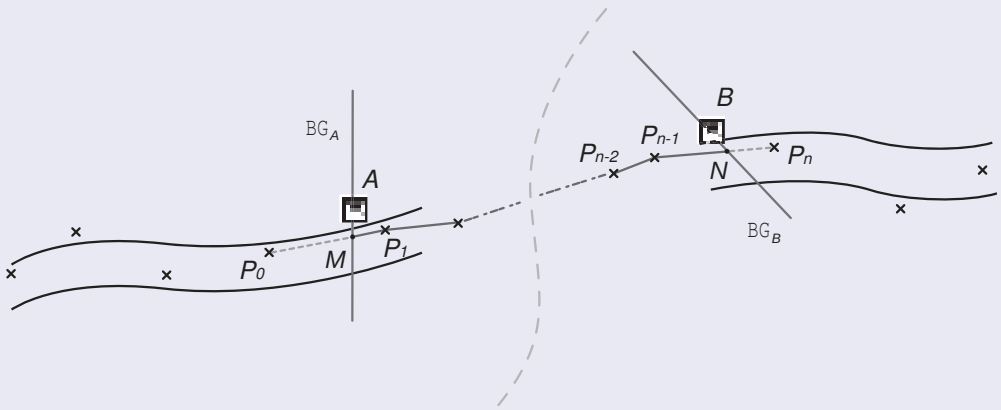




Journey extraction

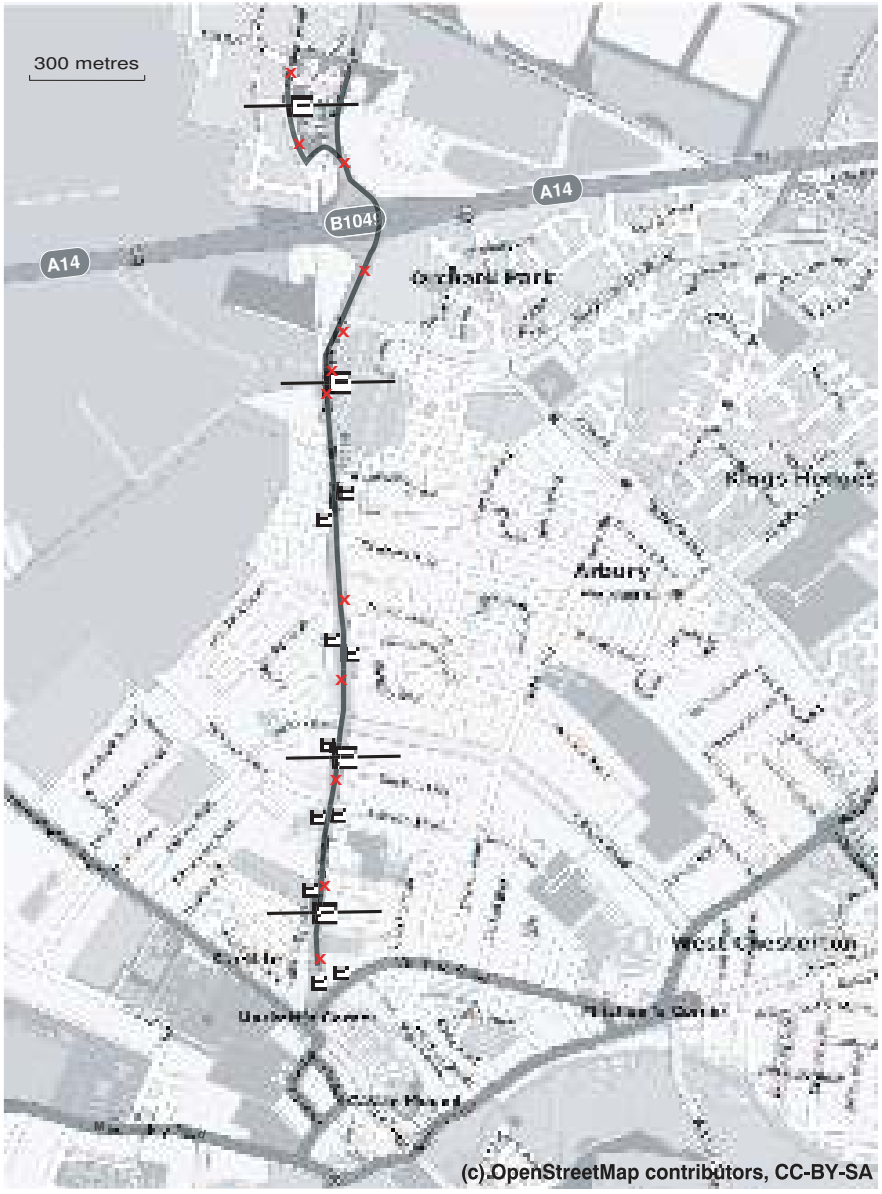
Methods:

- area-of-interest bounding rectangles (difficult to automate)
- bus stop gate technique: intersection of $\rho = P_0 P_1 \ldots P_n$ with bus stop gates defining a route $r_{AB} = A \rightsquigarrow B$

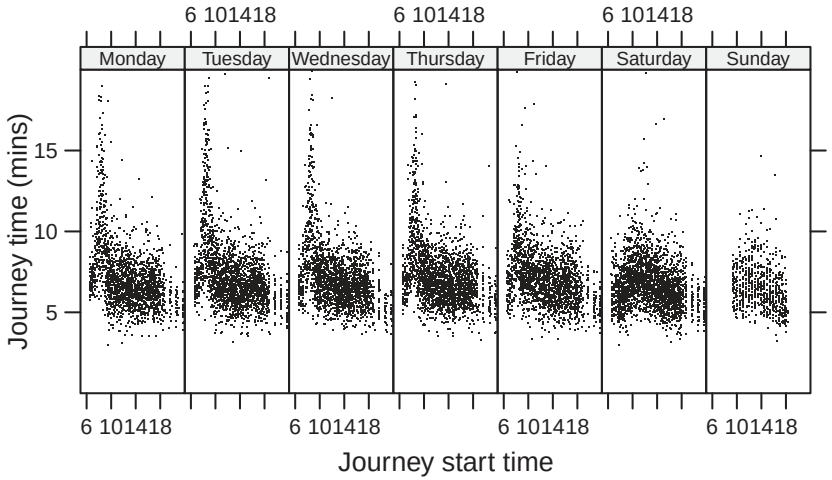


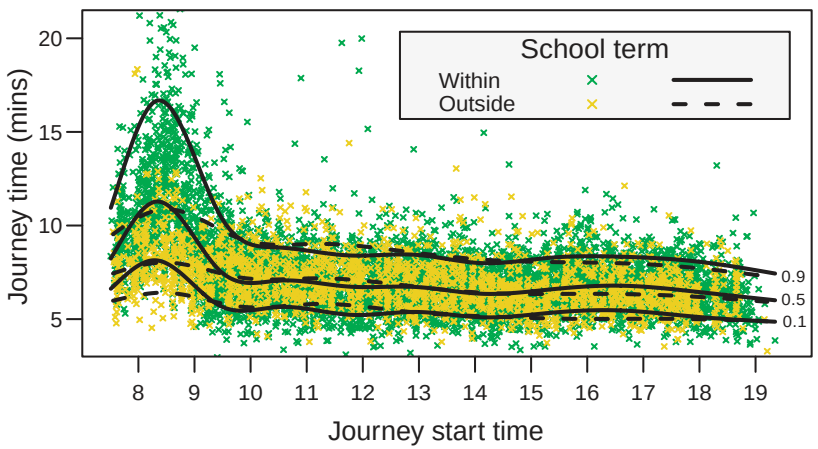
$$JT(\rho; r_{AB}) = t_n - t_0 - \left(\frac{P_0 M}{P_0 P_1}(t_1 - t_0) + \frac{N P_n}{P_{n-1} P_n}(t_n - t_{n-1}) \right).$$

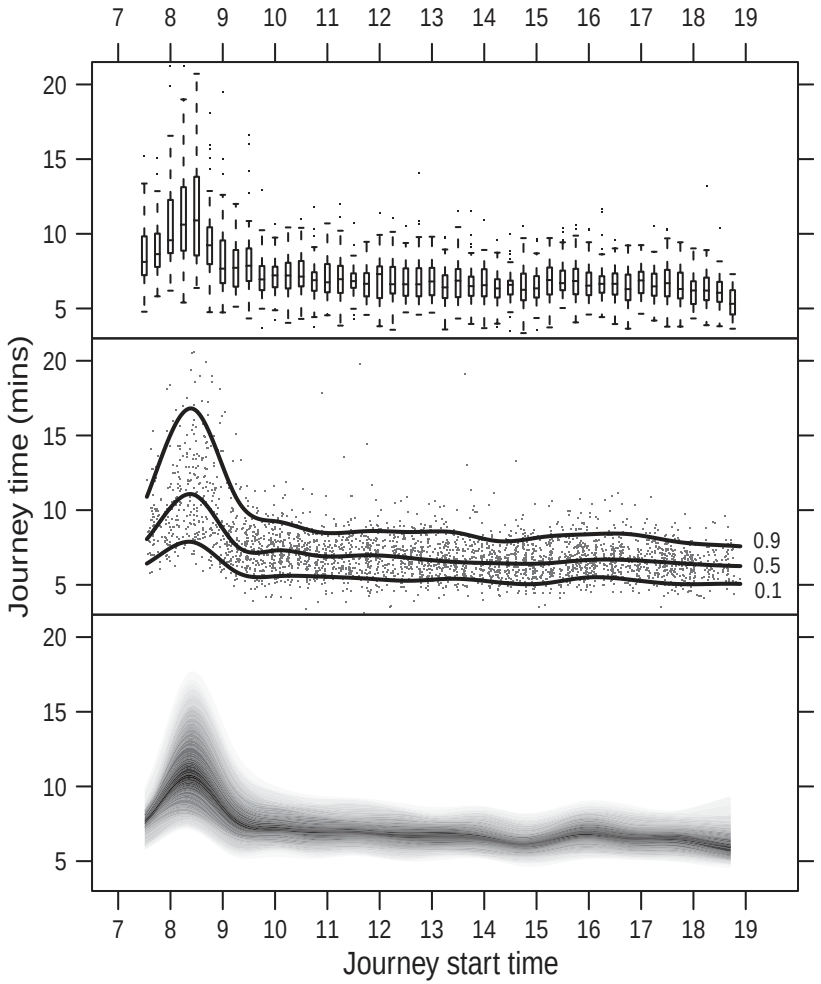
Journey time: Impington/Histon –> Histon Road –> Victoria Road



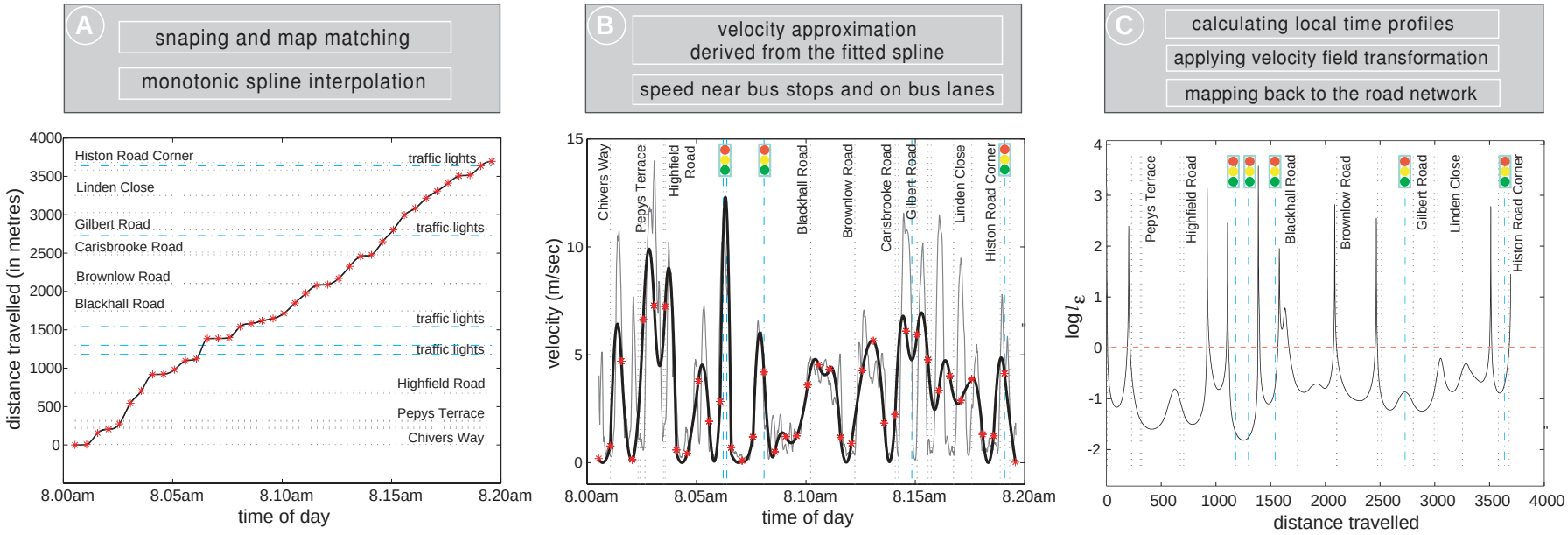
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Restoring behaviour of buses via spline interpolation



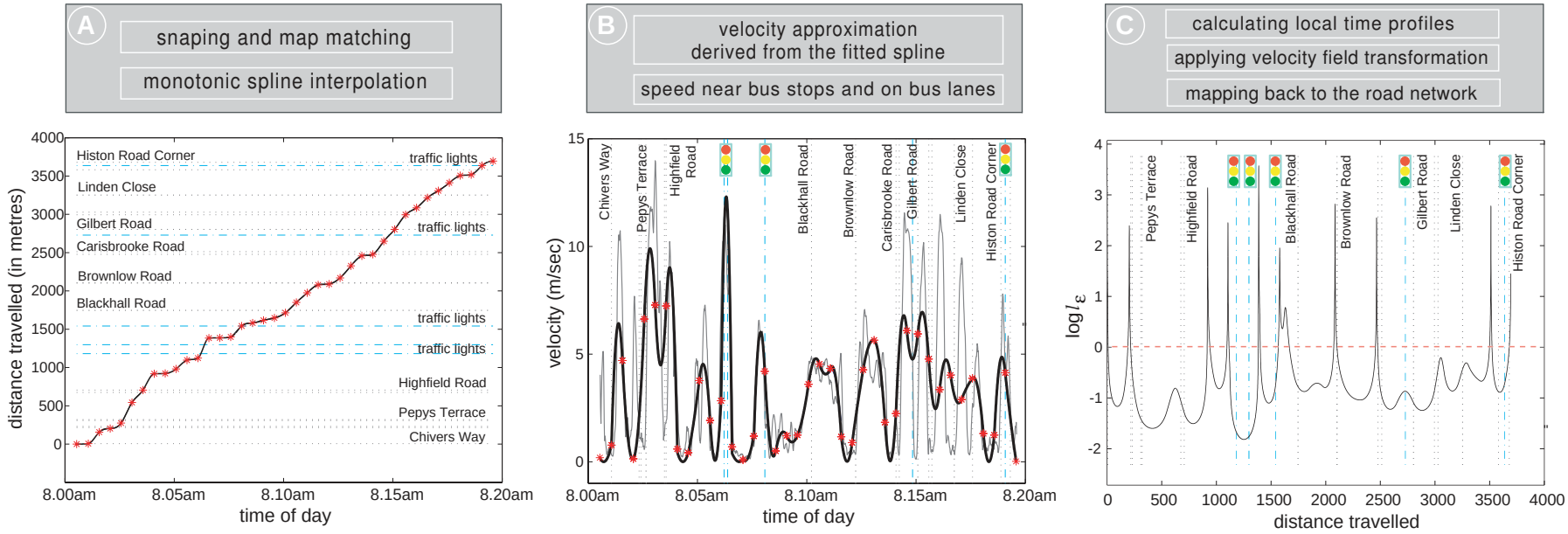
Time spent by a vehicle in ε-neighbourhood of a point s:

$$\ell_\varepsilon(s) = d^{-1}(s + \varepsilon/2) - d^{-1}(s - \varepsilon/2),$$

where d^{-1} is a generalised inverse function, defined as follows:

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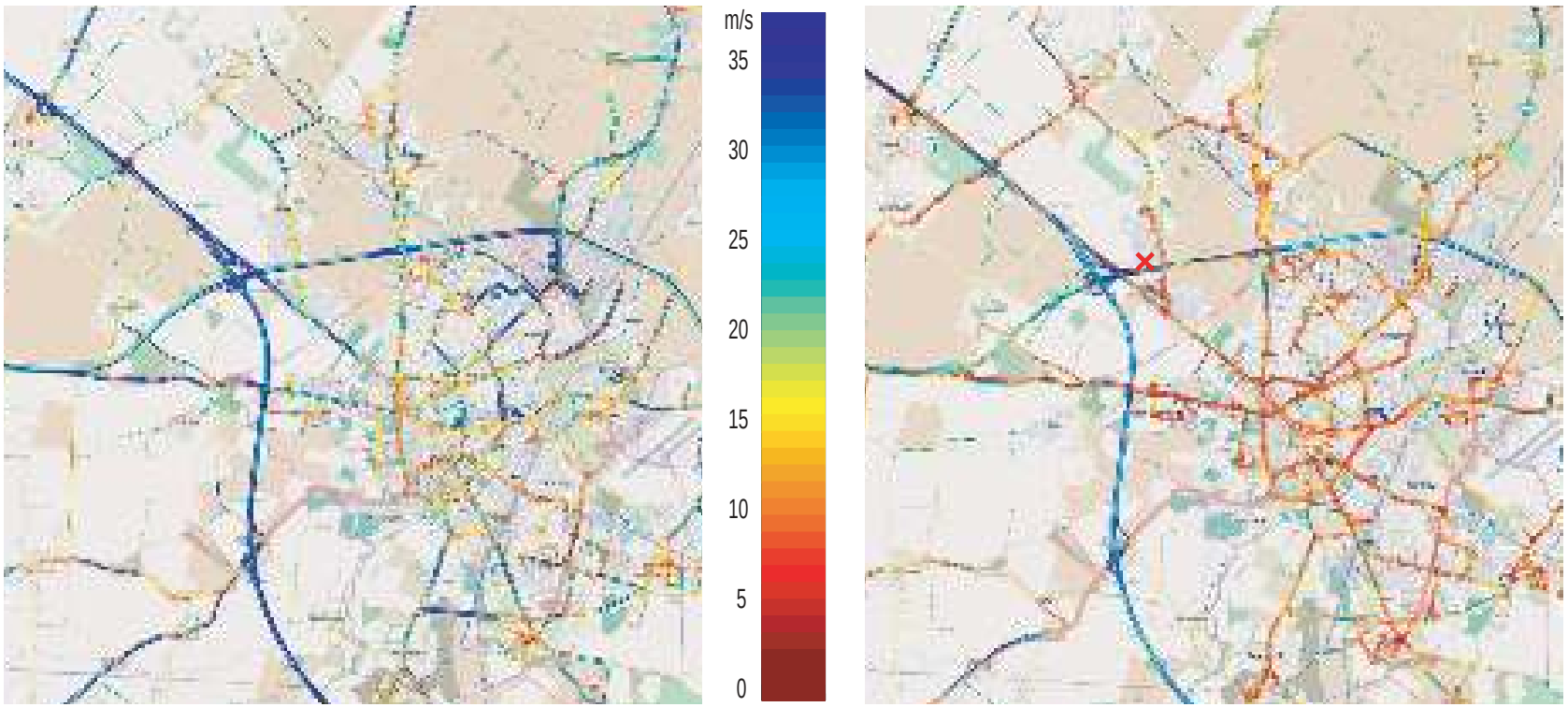
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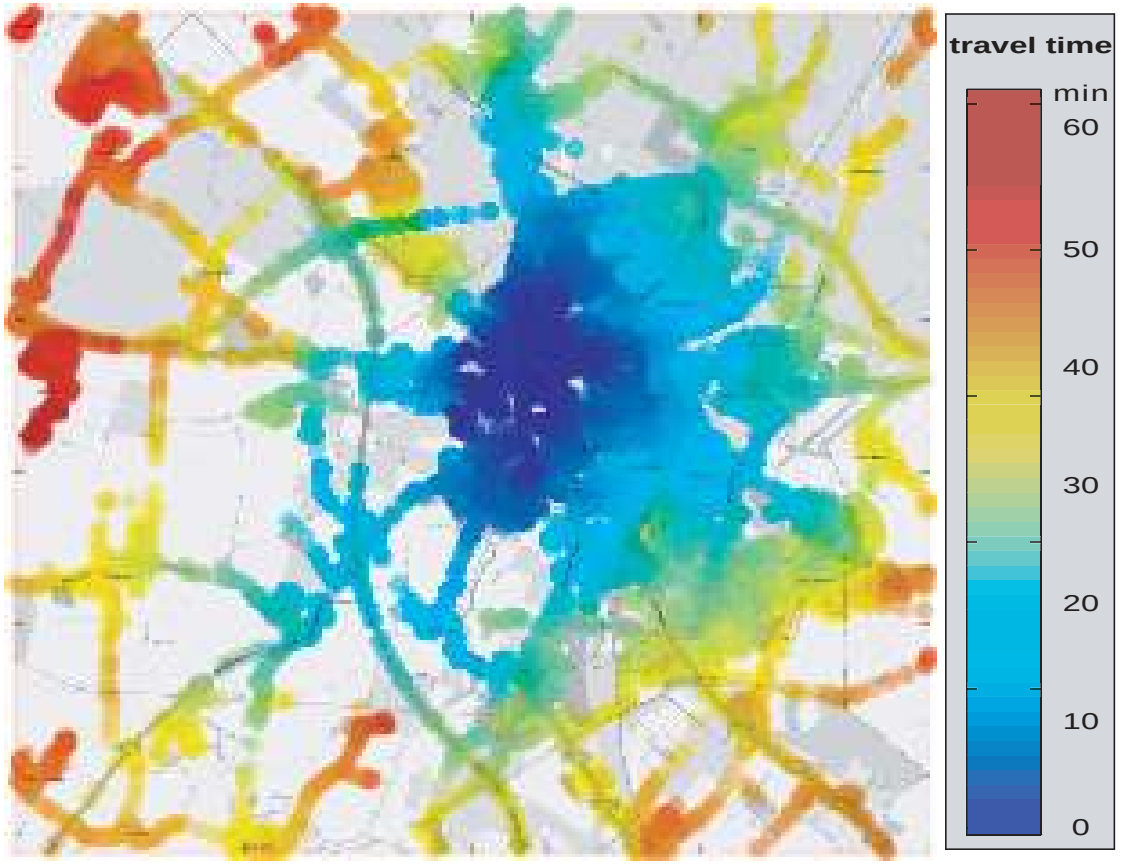
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Snapshot visualisations of the velocity fields

15 June and 22 June 2010 (A14 accident)



22 June 2010 (A14 accident)



Summary

We have presented effective techniques of using sparse bus data in studying travel times (of buses and general vehicles), road traffic dynamics and congestion. These techniques can be used by in RTI applications, to more optimally manage public transport fleets, and for better informativeness of the road users.

OpenStreetMap data were used in our analysis for the following purposes:

1. **Road network**: linear segment representation of the road network (for snapping and map-matching purposes).
2. **Bus stops**: locations and bearings of bus stops for journey extraction (bus stop gate technique).
3. **Bus routes**: description of bus routes permits to identify a subgraph of the road network giving a set of roads which are most likely to be used by buses; this subgraph helps to reduce uncertainty in snapping and map-matching process.
4. **Bus lanes, barriers and bollards**: parts of the road may be used by buses in an exclusive way.
5. **Maps**: these were used for illustration purposes.

Pitfalls: (in)consistency in tagging bus stops, slow or no reflection on temporary changes in bus routes, etc.

These techniques make use of OSM and therefore do depend on the

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STATE OF THE MAP!

References



Bacon J, Bejan A, Evans D, Gibbens R, Moody K (2011) Using Real-Time Road Traffic Data to Evaluate Congestion. *Lecture Notes in Computer Science* **6875**. Springer.



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Bejan A, Gibbens R, Evans D, Beresford A, Bacon J, Friday A (2010) Statistical Modelling and Analysis of Sparse Bus Probe Data in Urban Areas. *13th International IEEE Conference on Intelligent Transportation Systems*, Madeira Island, Portugal, IEEE Intelligent Transportation Systems Society (September 2010): 1256–1263.



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